

# UniLex Supporting Materials

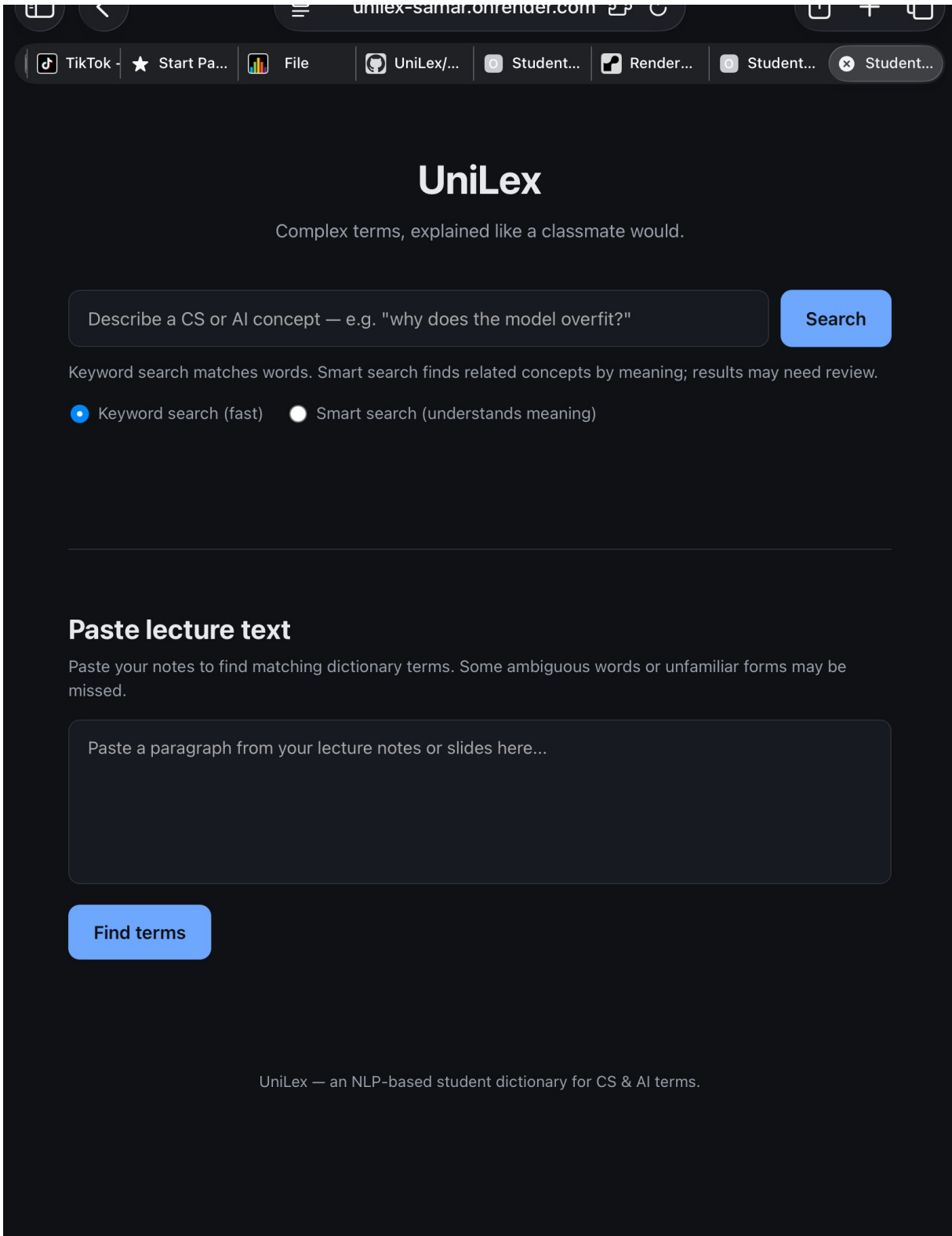
Website Screenshots and Feature-Testing Evidence

**Live Website:** <https://unilex-samar.onrender.com/>

**GitHub Repository:** <https://github.com/sara78606/UniLex>

This document presents screenshots of the UniLex interface and its main features. The descriptions explain what each screenshot demonstrates and provide supporting evidence for the M3 report.

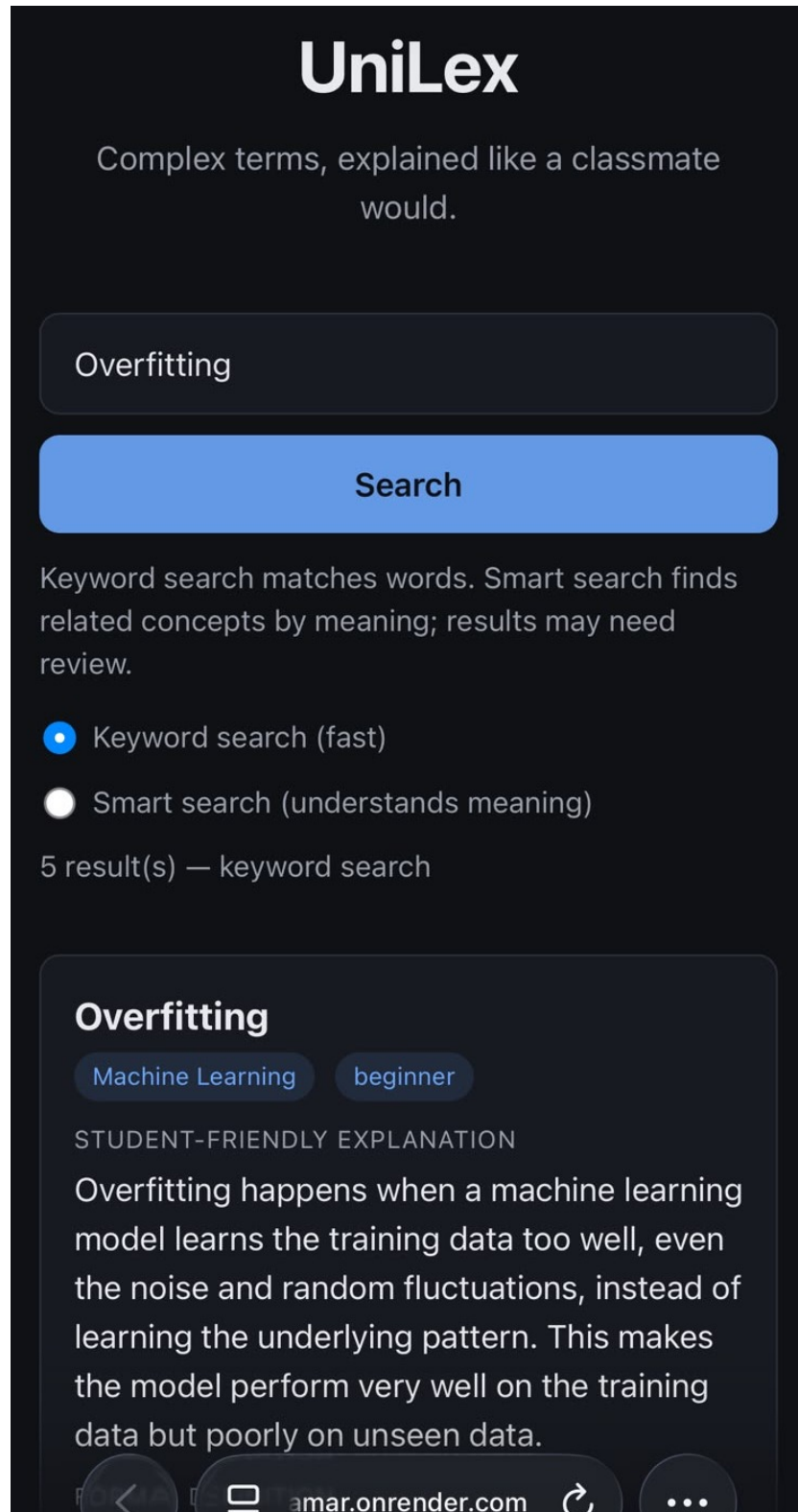
**Figure 1. UniLex Homepage**



*Figure 1. UniLex Homepage*

The homepage provides the main search field, options for keyword and smart search, and a separate section for extracting terms from lecture text. This screenshot shows the interface before a search is submitted.

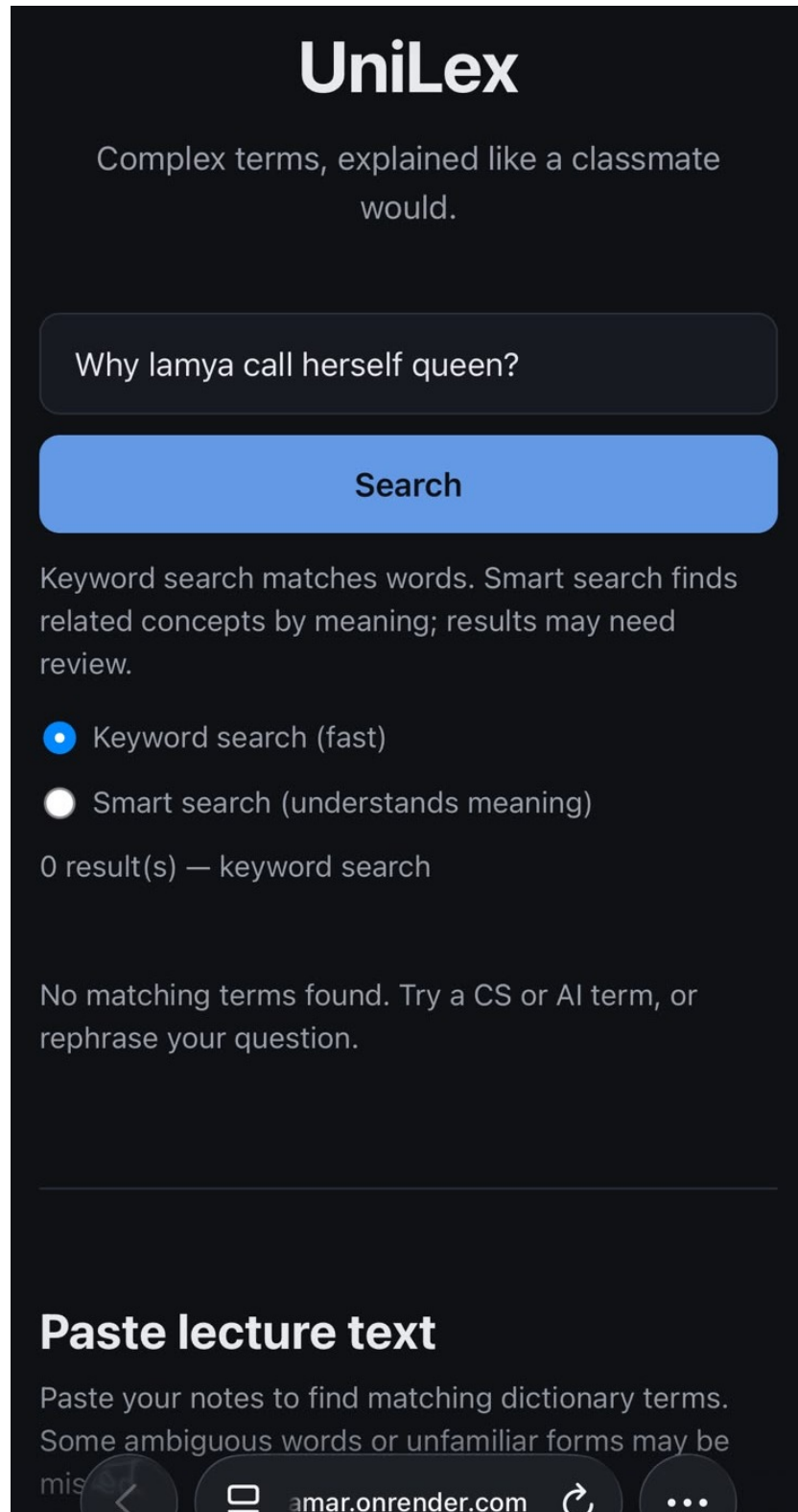
**Figure 2. Keyword Search – Overfitting**



*Figure 2. Keyword Search – Overfitting*

The keyword search returns results for the term “Overfitting”. The result includes a student-friendly explanation and identifies the concept as a beginner-level Machine Learning term.

**Figure 3. Keyword Search – No Matching Results**



*Figure 3. Keyword Search – No Matching Results*

This screenshot shows how UniLex responds when a query does not match the glossary. The website displays zero results and suggests trying a Computer Science or Artificial Intelligence term or rephrasing the query.

**Figure 4. Smart Search Result – AI Knowledge Graph Ontology**

**An Ai Knowledge Graph Ontology**

**Definition**

Artificial Intelligence intermediate

STUDENT-FRIENDLY EXPLANATION

An AI Knowledge Graph Ontology definition is the structural, overarching semantic blueprint and rule framework that dictates the allowed types of entities, properties, and directed relationships that can exist within a specific graph database schema. While a raw database holds facts, the ontology defines the logical laws of the world—specifying that a node classified as Person can have a relationship named works\_at pointing strictly to a node classified as Organization, but cannot point to an abstract numeric value. For example, an online healthcare knowledge graph uses an ontology to define strict medical logic boundaries connecting diseases, symptoms, and drug treatments. This matters because it forces unstructured data into a highly logical, consistent, and computer-readable format, maximizing the accuracy of AI context retrieval.

FORMAL DEFINITION

An AI Knowledge Graph Ontology is a structured, semantic blueprint and rule framework that defines the types of entities, properties, and relationships within a graph database schema.

EXAMPLE

A healthcare knowledge graph using an ontology to connect diseases, symptoms, and treatments.

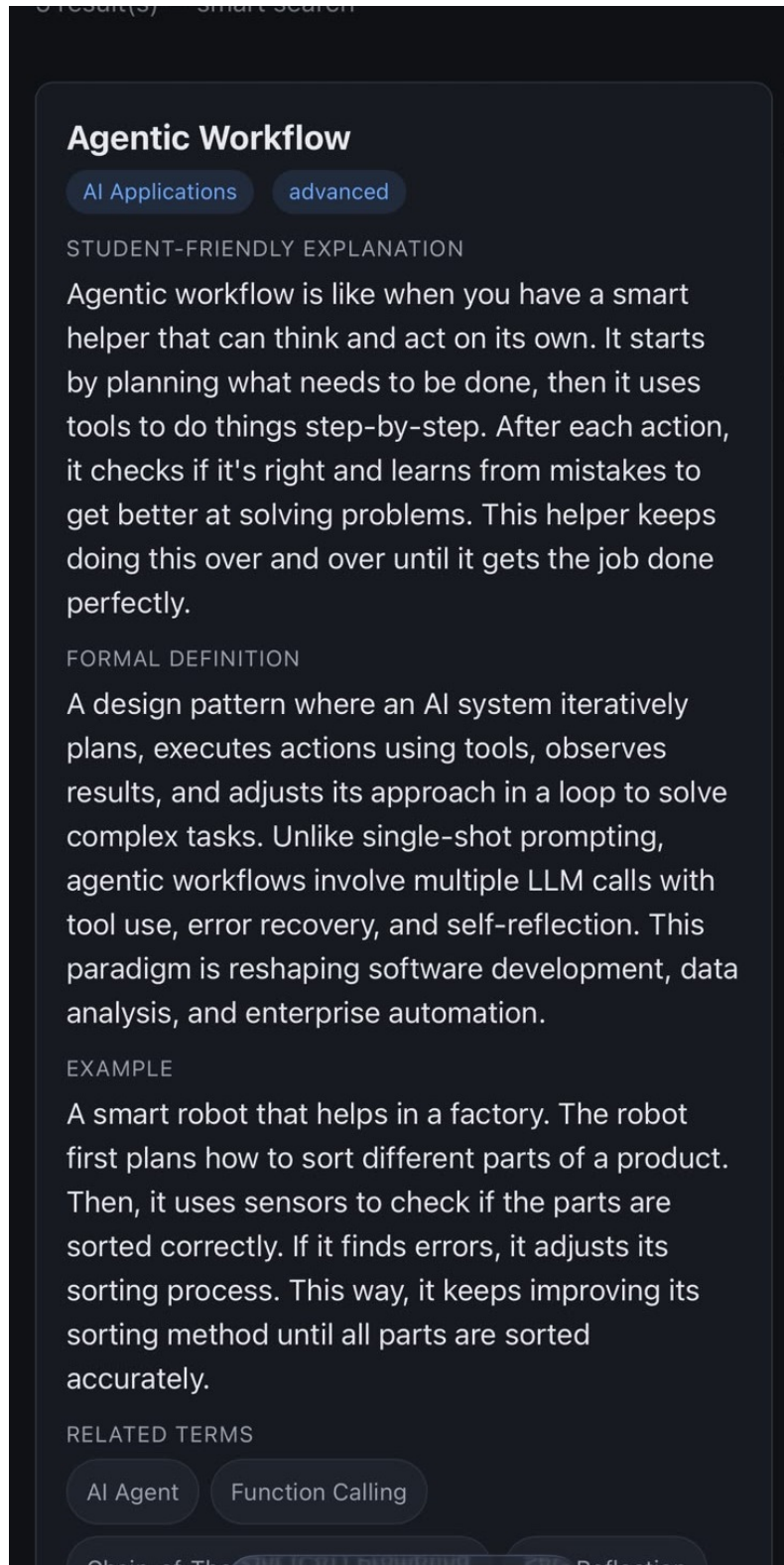
RELATED TERMS

Knowledge Graph Ontology Graph Database

*Figure 4. Smart Search Result – AI Knowledge Graph Ontology*

Smart search returns a related technical concept based on the meaning of the query. The result presents a student-friendly explanation, a formal definition, an example, and related terms.

**Figure 5. Smart Search Results – Agentic Workflow**



*Figure 5. Smart Search Results – Agentic Workflow*

This screenshot shows smart search returning results for a meaning-based query. The result card includes a simplified explanation, formal definition, example, and related terms. Search results should be reviewed for relevance.

**Figure 6. Lecture Text Extraction**

unilex-samar.onrender.com

## Paste lecture text

Paste your notes to find matching dictionary terms. Some ambiguous words or unfamiliar forms may be missed.

A machine learning model may suffer from overfitting when it performs well on training data but poorly on unseen data. Training data is used to learn model parameters, while testing data helps evaluate performance.

**Find terms**

3 term(s) found

### Machine Learning

Artificial Intelligence intermediate

STUDENT-FRIENDLY EXPLANATION

Machine Learning is like teaching a computer to learn from examples instead of following strict rules. It helps computers get better at tasks by learning from data.

FORMAL DEFINITION

A subfield of artificial intelligence in which systems learn patterns from data and improve their performance on a task without being explicitly programmed for every rule.

EXAMPLE

Google's spam filter uses Machine Learning to learn what emails are spam and what are not, improving its accuracy over time.

RELATED TERMS

AI Data Mining Deep Learning

SOURCE

wikipedia\_glossary

### Overfitting

Machine Learning beginner

STUDENT-FRIENDLY EXPLANATION

Overfitting happens when a machine learning model learns the training data too well, even the noise and random fluctuations, instead of learning the underlying pattern. This makes the model perform very well on the training data but poorly on unseen data.

FORMAL DEFINITION

A modeling error that occurs when a machine learning model learns the training data too well, including its noise and random fluctuations, resulting in poor generalization to new data. The model performs excellently on training data but poorly on validation or test data. Techniques

*Figure 6. Lecture Text Extraction*

The lecture-text feature identifies terms from pasted course material. In this example, UniLex detects three terms: Machine Learning, Overfitting, and related technical terminology available in the glossary.